

Md. Sohel

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PROFESSIONAL SUMMARY

Data-driven Environmental Science undergraduate with practical experience in data analysis, statistical interpretation, dashboard development, and research reporting. Skilled in GIS, Python, SQL, Power BI, and Advanced Excel for data cleaning, visualization, and decision-making support. Experienced in transforming complex datasets into actionable insights through statistical analysis, interactive dashboards, and research-based reporting. Strong analytical, problem-solving, and communication skills with a passion for leveraging data to support evidence-based decisions.

EDUCATION

2020	Secondary School Certificate (SSC) (Science Group) GD Shamsuddin Ahmed Collegiate School, Kumarkhali, Kushtia GPA 5 out of 5
2022	Higher Secondary Certificate (HSC) (Science Group) Kumarkhali Govt. College, Kumarkhali, Kushtia GPA 5 out of 5
2027	BSc in Environmental Science and Disaster Management Patuakhali Science and Technology University (PSTU) 2024 – 2028 (<i>Running</i>)

TECHNICAL SKILLS

Data Analysis & Statistics: Descriptive Statistics, Correlation Analysis, Survey Data Analysis, Data Cleaning and Transformation, Data Interpretation and Reporting

Programming & Databases: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL (Data Querying and Database Management), R (Basic)

Business Intelligence & Visualization: Power BI Dashboard Development, Advanced Microsoft Excel, Pivot Tables and Lookup Functions, Interactive Data Visualization

Research & Data Management: Data Collection and Validation, KoboToolbox Survey Management, Technical Report Writing, Research Methodology

GIS & Spatial Analytics: ArcGIS, QGIS, Google Earth Engine (GEE), PostGIS

PROJECTS

Spatiotemporal Forest Canopy Density Analysis of the Sundarbans

Performed a 30-year remote sensing–based assessment of Forest Canopy Density (1989–2020) in the Sundarbans. The project utilized multiple vegetation and environmental indices, including AVI, BSI, CSI, and SSI, to evaluate changes in forest health, canopy condition, and ecosystem stress, providing valuable insights for sustainable forest management and conservation planning.

GIS-Based Transportation Corridor Risk Assessment Using AHP-MCDM

Conducted a GIS-based transportation corridor risk assessment by integrating climatic, environmental, and socioeconomic factors within an AHP-MCDM framework. The study developed weighted vulnerability models and spatial risk maps to identify high-risk transportation corridors and support infrastructure resilience and strategic planning.

Comprehensive Earthquake Risk Assessment and Vulnerability Mapping of Dhaka City (Ongoing)

This ongoing research applies GIS and spatial analysis techniques to evaluate earthquake risk and urban vulnerability in Dhaka City. The study examines seismic hazards, structural exposure, evacuation challenges, and urban resilience factors to support disaster preparedness and evidence-based urban risk management.

RESEARCH AND PUBLICATIONS

Preprint Article: Development of a GIS-Based Household Vulnerability Index and Micro-Trigger System for Anticipatory Cyclone Action in Patharghata Union, Barguna, Bangladesh.

CERTIFICATIONS

- *Craft Effective Prompts for Microsoft 365 Copilot* – Microsoft

- *Going place with spatial analysis* - ESRI
- *Microsoft Office* – EDGE Course
- *Data Science Landscape* – Dataskool
- *COSHH Manager Certification* – Knights of Safety

KEY ACHIEVEMENTS

First Place Winner – Awarded 1st Place at the "Bridging Youth and Opportunity: Conference on Anticipatory Action" for developing an innovative project proposal development.

LEADERSHIP & ACTIVITIES

Founder – Career Portal

Executive Member – ESDM Club PSTU

General Member – PSTU Research Society

SOFT SKILLS

Analytical Thinking, Problem Solving, Data Storytelling, Communication Skills, Teamwork, Collaboration, Time Management, and Adaptability

LANGUAGES

Bengali – Native

English – Professional Working Proficiency

CAREER INTERESTS

Research Analytics | GIS Developer | Evidence-Based Decision Maker